

MOBILITY PREDICTORS:

Identifying Structural Conditions for Upward Economic Mobility

Jeff Charles

DSE6311: Capstone, Merrimack College

May 2026

Submitted to: Policymakers and Nonprofit Organizations Working on Economic Mobility and

Housing Equity

Including the Urban Institute and the National Low Income Housing Coalition (NLIHC)

GitHub: <https://github.com/JCharles8/mobility-predictors>

Synopsis

For every 100 low-income households in America in 2010, fewer than 22 reached the middle class by 2021. This project identifies why age, race, and education are the dominant structural barriers, with White households more than twice as likely to achieve mobility as Black households, even after controlling for wealth, income, and housing costs. Asset-building programs, rental cost relief, and race-conscious policy design carry the greatest leverage for organizations working to close that gap.

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Research Question

Among households in the bottom 40 percent of the U.S. income distribution in 2010, which combination of structural baseline conditions, including wealth, housing cost burden, family structure, educational attainment, employment status, geography, and demographic characteristics, best predicts whether a household crosses the national median income threshold by 2021?

Background

The American ideal of upward mobility has eroded markedly over the past half-century. Chetty et al. (2017) document that approximately 90 percent of children born in 1940 went on to out-earn their parents, while that figure had fallen to roughly 50 percent for the cohort born in 1980. For households already at the bottom of the income distribution, the population examined in this project, the challenge is more acute. The Panel Study of Income Dynamics (PSID), the longest-running longitudinal household survey in the United States, provides a uniquely powerful lens for studying these dynamics because it tracks the same families across time, capturing how income, wealth, housing, employment, and demographic characteristics at one point in life predict outcomes a decade later.

Two bodies of evidence motivate the specific hypotheses in this project. First, Chetty et al. (2014) demonstrate that neighborhood-level structural conditions, including racial composition, income inequality, and local institutional quality, are stronger predictors of intergenerational income mobility than individual characteristics like parental education or earnings alone. Second, Desmond (2016) documents, through detailed ethnographic and quantitative work, that housing cost burden functions as a structural

trap for low-income households: when housing consumes more than 30 percent of income, the capacity to save, invest in education, or absorb unexpected expenses is severely constrained. Both findings suggest that structural factors at baseline, particularly wealth and housing burden, may carry more predictive weight for mobility than educational attainment, which is the conventional lever emphasized in workforce development and social mobility policy.

This project is addressed to policymakers and nonprofit organizations working on economic mobility and housing equity, including the Urban Institute and the National Low Income Housing Coalition (NLIHC), as primary stakeholders. Its purpose is to produce a ranked predictor model that these organizations can use to understand which structural conditions carry the greatest leverage for program design, policy advocacy, and resource allocation.

Hypothesis and Predictions

The central hypothesis is that structural wealth and housing cost burden at baseline are stronger predictors of upward income mobility than educational attainment alone, and that their predictive power compounds across demographic groups. Specifically:

1. Household net worth at baseline will rank among the top three most important features in both Random Forest and XGBoost models, outperforming educational attainment.
2. Housing cost burden will be negatively associated with mobility outcomes and carry stronger predictive power than geographic region alone.

3. K-Means clustering will identify a distinct high-burden, low-asset household profile with substantially lower mobility rates than other clusters.
4. Educational attainment will retain predictive value but will be conditional on baseline wealth, reflecting findings that credentials amplify existing structural advantage rather than substituting for it (Putnam, 2015).

As described in the Results section, Predictions 2 and 3 are confirmed. Prediction 1 is partially confirmed: net worth ranks consistently in the top six predictors across all models, but education outranks it in tree-based importance rankings. Prediction 4 is partially disconfirmed by the SHAP interaction analysis, which finds education and wealth operating as roughly independent contributors rather than in a strongly conditional pattern.

Data

Data Acquisition

The primary dataset is the Panel Study of Income Dynamics (PSID), maintained by the Institute for Social Research at the University of Michigan (University of Michigan Institute for Social Research, 2023). The PSID is the longest-running longitudinal household survey in the United States, launched in 1968 with a nationally representative sample of approximately 5,000 households and collected biennially since 1997. It captures income, wealth, employment, housing, demographics, and family structure across waves, making it uniquely suited to mobility research requiring baseline and follow-up observations for the same household.

Three files were used: the 2011 Family File (FAM2011ER), the 2021 Family File (FAM2021ER), and the Cross-Year Individual File (IND2023ER). All files were obtained

as fixed-width ASCII text files from the PSID public-use data portal at psidonline.isr.umich.edu, which requires free researcher registration. No API or web scraping was used. The 2011 wave is the baseline because it aligns with a PSID wealth supplement wave, making household net worth data directly available. The 2021 wave provides the follow-up income outcome. The Cross-Year Individual File provides person-level panel identifiers to track the same household head across waves.

The full 2011 family file contains 8,907 household family units; the 2021 family file contains 9,207; and the Cross-Year Individual file contains 85,536 individual records spanning 1968 to 2023. The panel-linked analytic sample restricted to the bottom 40 percent of the 2010 income distribution contains 2,375 households, the dataset used for all analysis in this report.

The analytic sample is well-matched to the research question. The income restriction ensures that all households face similar resource constraints at baseline, making differences in structural conditions, wealth, housing cost burden, education, and geography the primary variation of interest rather than income-level differences. PSID's longitudinal design is essential for measuring mobility: by tracking the same household from 2011 to 2021, income changes reflect genuine household trajectories rather than cross-sectional compositional differences. One data collection caveat: the PSID oversamples low-income households, which improves statistical power for this study but means that mobility rate estimates are not representative of the full U.S. population. The data dictionary is provided in Appendix A.

Data Cleaning and Panel Linkage

Variable extraction was performed in Python by parsing the column-position specifications in the Stata setup files. The panel linkage proceeded in three steps. First, the Cross-Year Individual file was used to identify individuals who served as household head (sequence number = 1) in the 2011 wave and who had a valid 2021 interview number, yielding 5,998 linked head-level records. Second, this linkage file was merged with the 2011 Family File on the 2011 interview number and with the 2021 Family File on the 2021 interview number, producing a panel-linked dataset of 5,998 households. Third, the sample was restricted to households at or below the 40th percentile of the 2010 income distribution (\$37,190), yielding the final analytic sample of 2,375 households.

Several variables required recoding. Educational attainment was grouped into five ordered categories: Less than High School, High School Diploma, Some College or Associate's Degree, Bachelor's Degree, and Graduate Degree. Employment status was consolidated from eight original codes into five categories. Race and ethnicity were combined from separate race and Hispanic origin variables into four categories, with Hispanic origin taking precedence. Geographic region was mapped from PSID Census region codes to four standard U.S. regions. Housing cost burden was computed as the ratio of annualized monthly housing expenditure to total family income and is missing for 606 households in "Neither" tenure arrangements. PSID-specific missing value codes were recoded to system-missing before any analysis.

The target variable (`above_median_2021`) is a binary indicator of whether the household's 2020 income, reported in the 2021 PSID wave, exceeded the national

median of \$67,521. Of the 2,375 households in the analytic sample, 21.5 percent achieved this threshold.

Data Exploration

Descriptive Statistics

Table 1 presents descriptive statistics for all continuous predictor variables. The most analytically striking finding is the contrast between income and wealth distributions. Household income is approximately symmetric around a median of \$19,554, expected given the income-based sample restriction. Wealth is dramatically more unequal: median net worth excluding home equity is only \$1,000, while the mean is \$17,952, driven by a small number of households with substantially higher assets. At least one in four households had zero or negative net worth at baseline, meaning a single unexpected expense, a medical bill, or a car repair, could immediately destabilize their financial position with no buffer.

Variable	N	Mean	Median	SD	Min / Max	Skew / Kurt.
Household Income (2010, \$)	2,375	\$19,590	\$19,554	\$10,267	\$1 / \$37,190	-0.06 / -1.05
Net Worth Excl. Home Equity (\$)	2,375	\$17,952	\$1,000	\$142,423	-\$990K / \$9.5M	16.9 / 390.2
Net Worth Incl. Home Equity (\$)	2,375	\$36,211	\$2,000	\$181,458	-\$990K / \$10M	12.7 / 225.6
Age of Household Head (years)	2,375	40.0	36	15.7	17 / 96	0.71 / -0.42
Household Size (persons)	2,375	2.27	2	1.48	1 / 14	1.43 / 2.70
Number of Children	2,375	0.81	0	1.21	0 / 10	1.90 / 4.39

Variable	N	Mean	Median	SD	Min / Max	Skew / Kurt.
Monthly Housing Cost (\$)	1,769	\$542	\$500	\$374	\$1 / \$4,273	3.26 / 17.5
Housing Cost Burden (ratio)	1,769	0.432	0.288	0.429	0.00 / 2.00	2.41 / 5.39
Education of Head (years)	2,338	12.4	12	2.3	1 / 17	-0.60 / 2.17

Table 1. Descriptive statistics for all continuous predictor variables among panel-linked households in the bottom 40% of the 2010 income distribution (N=2,375). Housing cost statistics reflect the 1,769 households with available cost data. Extreme skewness and kurtosis for net worth variables reflect the highly right-skewed wealth distribution characteristic of low-income populations.

Table 2 presents frequency distributions for categorical predictors. Educational attainment is concentrated at the lower end: 60 percent of heads have no post-secondary education. Black and African American households represent 54.7 percent of the sample, a substantial overrepresentation relative to their approximately 13 percent share of the national population, consistent with documented racial disparities in income distribution. Geographically, 50.6 percent of households reside in the South. Among households with available housing cost data, 35.7 percent of the full analytic sample are cost-burdened at or above the HUD 30 percent threshold.

Variable	Category	N	Percent
Educational Attainment	Less than High School	604	25.8%
	High School Diploma	799	34.2%
	Some College / Associate's	538	23.0%
	Bachelor's Degree	100	4.3%
	Graduate Degree	297	12.7%
Employment Status	Working	1,311	55.2%
	Temporarily Laid Off	18	0.8%

Variable	Category	N	Percent
	Unemployed	437	18.4%
	Not in Labor Force	406	17.1%
	Other	201	8.5%
Family Structure	Married / Partnered	1,770	74.5%
	Single / Unpartnered	605	25.5%
Race / Ethnicity	Black / African American	1,289	54.7%
	White (Non-Hispanic)	889	37.7%
	Hispanic	100	4.2%
	Other	77	3.3%
Census Region	South	1,195	50.6%
	Midwest	600	25.4%
	West	348	14.7%
	Northeast	218	9.2%
Housing Tenure	Renter	1,499	63.1%
	Owner	613	25.8%
	Neither (incl. free housing)	263	11.1%
Cost Burden Category	Not Cost Burdened (<30%)	920	38.7%
	Cost Burdened (≥30%)	849	35.7%
	Housing Cost Not Reported	606	25.5%

Table 2. Frequency distributions for all categorical predictor variables in the panel-linked analytic sample (N=2,375). Cost burden percentages reflect proportions of the full analytic sample.

Key EDA Visualizations and Findings

Nine visualizations were produced during the EDA phase. The five most analytically consequential findings are summarized here.

The wealth distribution (Figure 2 in the EDA report) reveals the central structural diagnostic for this project: at least one in four households had zero or negative net worth at baseline. The median net worth excluding home equity is only \$1,000. For policymakers at NLIHC, this wealth floor is the key finding: two households earning identical incomes face fundamentally different risk profiles and mobility prospects depending on whether they have \$0 or \$50,000 in assets behind them.

The housing cost burden distribution (Figure 3) shows that among households with reported costs; 48 percent are cost-burdened at or above the HUD 30 percent threshold. The mean burden ratio (0.432) is pulled substantially higher than the median (0.288) by households spending more than half their income on housing.

The observed mobility rate by structural predictor (Figure 9) provides the most direct preview of the project's central findings. The gradient by educational attainment spans from 10.9 percent (less than high school) to 48.5 percent (graduate degree). White (Non-Hispanic) households achieved mobility at 33.2 percent versus 13.3 percent for Black/African American households, a 2.5x gap that, by sample construction, cannot be explained by income differences.

The correlation matrix (Figure 8) identified two primary collinearity concerns. Net worth excluding and including home equity are highly correlated ($r = 0.89$); only the excluding-equity measure was retained for models. Housing cost burden is mechanically correlated with income ($r = -0.22$) because it is computed as housing cost divided by income; the components were entered separately to address this.

The bivariate wealth-by-education analysis (Figure 6) revealed a clear positive gradient: graduate-degree holders have substantially higher median wealth even within the low-income sample. This pattern supports Prediction 4, that education and wealth are intertwined, but also confirms that wide intra-group variance in wealth exists at every education level.

Models

Preprocessing and Feature Engineering

Train-Test Split

The 2,375-household analytic dataset was split 80/20 into training ($n = 1,898$) and test ($n = 475$) sets using stratified random assignment. Stratification was performed jointly on age quartile and the binary target to preserve the 21.5 percent positive class rate across both sets AND to prevent cohort composition bias, a risk when younger households would otherwise be concentrated in the training set and produce age-conditioned patterns that do not generalize to older test-set households. All preprocessing objects were fit exclusively on training data; no test-set information was used during any fitting step.

Missing Data

Two variables required imputation. Annual housing cost is missing for 606 households (25.5 percent) in "Neither" tenure arrangements. Education years is missing for 37 households. Before imputation, a binary indicator (`housing_cost_missing`) was constructed for the housing cost variable to preserve the structural signal embedded in the Neither tenure category, households with employer-provided or family-provided

housing, have systematically different economic circumstances than renters or owners, and that signal would be lost if imputed first. Multiple Imputation by Chained Equations (MICE), implemented via scikit-learn's `IterativeImputer` with `max_iter=10` and `random_state=42`, was applied to both variables on the training set only, then applied to the test set via `transform` (no `refit`).

Feature Scaling

`StandardScaler` was applied to all continuous features, fit on the training set only. Logistic regression requires scaling for coefficient stability; tree-based models are scale-invariant but receive the same input for pipeline consistency.

Wealth Transformation

A sign-preserving log transformation was applied to net worth. The raw wealth variable has extreme right skewness ($\text{skew} = 16.9$) and includes negative values (households with negative net worth), which make standard $\log_{1p}$ transformation inapplicable. The custom `sign_preserving_log` function computes $\text{sign}(x) * \log_{1p}(|x|)$, preserving the direction of negative net worth while compressing the magnitude. Post-transformation skewness was reduced from approximately 17.0 to approximately 4.5, a substantial improvement, though residual skewness remained.

K-Means Clustering

K-Means clustering ($k=3$) was applied as a feature engineering step to capture latent household typologies that the individual features do not fully represent. The clustering was performed on six structural features (log net worth, annual housing cost, baseline income, age, household size, and education years) after standardization. K was selected

based on elbow method analysis and silhouette scores across $k = 2$ through 6 ; $k = 3$ produced the clearest structural differentiation. Three household profiles emerged:

- Cluster 0 (High Burden / Low Asset): 14.9% mobility rate, low net worth, high housing cost relative to income, predominantly renters.
- Cluster 1 (Moderate Wealth): 28.0% mobility rate, moderate net worth, manageable housing costs, mix of owners and renters.
- Cluster 2 (Low Cost / Neither Tenure): 25.4% mobility rate, housed through employer or family arrangements, low-cost burden but limited asset accumulation.

The K-Means model was fit on training data only. Test observations were assigned to clusters via `predict()` with no refit on test data. A leakage audit was conducted by training a Random Forest classifier on `cluster_label` and `housing_cost_missing` alone and evaluating AUC on the test set; the combined audit AUC was 0.61, and `cluster_label` alone produced an AUC of 0.58. These values fall in the range expected for genuine structural signal (0.55 to 0.65) and confirm that the cluster feature is capturing real economic typologies rather than test-set contamination.

Algorithm Selection

Three supervised classification algorithms were used in sequence, moving from the most interpretable to the most powerful.

Logistic Regression (Baseline)

Logistic regression was selected as the baseline model for three reasons. First, it provides interpretable coefficient estimates that map directly to the research question: which structural predictors independently associate with mobility, and in what direction? Second, the linear decision boundary assumption is testable and explicitly addressed. Third, it provides a performance floor against which the complexity of ensemble models can be evaluated; if ensembles do not meaningfully outperform logistic regression, it suggests the key relationships are approximately linear, which is itself a finding. An L2 (Ridge) penalty was applied because the goal at baseline is coefficient interpretation rather than variable selection: L2 shrinks all coefficients proportionally rather than zeroing some out, preserving all features for interpretation. The class imbalance (21.5% positive) was addressed via `class_weight='balanced'`.

Logistic regression assumes: (1) linearity between log-odds and continuous predictors; (2) independence of observations; (3) absence of severe multicollinearity; and (4) no extreme outliers. Box-Tidwell tests found no statistically significant departure from linearity for education years and age head ($p > 0.10$). Log net worth and annual housing cost showed marginal non-linearity ($p \approx 0.04$), addressed by the sign-preserving log transformation. Variance Inflation Factors (VIF) for all continuous predictors were below 5.0, with the highest at 3.1 for annual housing cost. Cook's distance exceeded the $4/n$ threshold for zero observations; 12 observations exceeded $1/\sqrt{n}$, all associated with high-wealth households in the upper tail.

Random Forest

Random Forest was selected for its robustness to multicollinearity, built-in out-of-bag (OOB) error estimation, and Gini impurity importance scores that complement the

logistic regression coefficients. An OOB error curve across $n = 50$ to 500 confirmed stability around $n = 250$ (OOB error = 0.1828). Hyperparameter tuning used `RandomizedSearchCV` with $n_iter = 50$ and 5-fold stratified cross-validation, searching across $n_estimators$ (100 to 400), max_depth (3 to None), $max_features$ (sqrt, log2, 0.3, 0.5), $min_samples_leaf$ (1 to 10), and $min_samples_split$ (2 to 10). Class imbalance was addressed via `class_weight='balanced'`. `RandomizedSearchCV` was chosen over full `GridSearchCV` because the parameter space would require thousands of fits; 50 random combinations provide broad coverage at a fraction of the cost.

XGBoost

XGBoost was selected as the most powerful model in the sequence because gradient boosting captures additive nonlinear interactions that Random Forest's bagging approach does not directly model, and because it provides gain-based feature importance that weights splits by prediction quality rather than frequency. Class imbalance was addressed via `scale_pos_weight = 3.64` (the ratio of negative to positive class observations), XGBoost's native equivalent of balanced weighting. Hyperparameter tuning used the same `RandomizedSearchCV` setup ($n_iter = 50$, 5-fold CV, `roc_auc` scoring), searching across $n_estimators$ (100 to 500), max_depth (3 to 6), $learning_rate$ (0.01 to 0.2), $subsample$ (0.6 to 1.0), $colsample_bytree$ (0.6 to 1.0), min_child_weight (1 to 5), and $gamma$ (0 to 0.3). Learning rate and $n_estimators$ were tuned together because low learning rates require more trees to converge.

Final Model and Cross-Validation Results

Model Comparison

Table 3 presents the cross-validated and test set performance of all three models.

Model	CV AUC	Test AUC	CV-Test Gap	Best Parameters (abbreviated)
Logistic Regression	0.794	0.805	+0.011	C=1.0, L2
Random Forest	0.806	0.799	-0.007	n=300, depth=5, leaf=10, feat=0.3
XGBoost	0.811	0.795	-0.015	n=500, depth=3, lr=0.01, sub=0.8, $\gamma=0.1$

Table 3. Model performance comparison across three supervised classifiers. CV metrics from 5-fold stratified cross-validation on the training set ($n=1,898$). Test AUC from the held-out 20% split ($n=475$). All CV-test gaps are within ± 0.015 , confirming no meaningful overfitting across any model.

XGBoost achieves the best cross-validated AUC (0.811) and is designated the final model on that basis. ROC-AUC is the primary evaluation metric because it is threshold-independent and robust to class imbalance; accuracy is treated as secondary. The honest interpretation of the model comparison is important to state explicitly: the performance spread across all three models is only 17 points of AUC on CV and 10 points on the test set. The structural predictors in this dataset have approximately linear relationships with mobility outcomes, and the complexity of ensemble models provides modest incremental performance over the interpretable baseline. This is itself a finding; it simplifies the story for a policy audience and means the logistic regression coefficients offer a trustworthy portrait of the direction and relative magnitude of each predictor's association.

Table 4 presents classification metrics on the test set for the mobility class.

Model	Precision	Recall	F1	Accuracy
Logistic Regression	0.41	0.73	0.52	0.72
Random Forest	0.45	0.70	0.55	0.75
XGBoost	0.43	0.73	0.54	0.73

Table 4. Classification metrics on the held-out test set for the positive (mobility) class. All models use class balancing to optimize recall over precision, appropriate for a policy application where missing a mobile household is more costly than a false alert.

Overfitting Assessment

No model shows meaningful overfitting. CV-test gaps range from -0.015 to +0.011, all well within acceptable bounds for a training set of approximately 1,900 observations. The logistic regression's positive gap (+0.011) indicates it generalizes slightly better to the held-out test set than the CV folds suggest, confirming the L2 regularization is working appropriately. The XGBoost gap (-0.015) reflects the slightly higher complexity of the gradient boosting model; a gap of this magnitude is expected and not a concern for a dataset of this size.

Feature Importance

Table 5 presents the top 10 feature importances from Random Forest (Gini impurity) and XGBoost (gain), alongside the logistic regression coefficient rankings for comparison.

Rank	Random Forest	RF Importance	XGBoost	XGB Importance
1	age_head	0.329	age_head	0.106
2	race: White (Non-Hispanic)	0.112	race: White (Non-Hispanic)	0.100
3	income_2010	0.107	education_years	0.063
4	education_years	0.097	cluster_label	0.060
5	employment: Working	0.062	education: Graduate Degree	0.058
6	log_net_worth	0.061	employment: Working	0.058
7	annual_housing_cost	0.046	income_2010	0.053

Rank	Random Forest	RF Importance	XGBoost	XGB Importance
8	education: Graduate Degree	0.041	log_net_worth	0.037
9	cluster_label	0.030	hh_size	0.034
10	hh_size	0.025	annual_housing_cost	0.031

Table 5. Top 10 feature importances from Random Forest (Gini impurity) and XGBoost (gain). Rankings are consistent with logistic regression coefficient magnitudes, confirming that the patterns are not artifacts of any single model type.

Three findings stand out from Table 5. First, age of household head ranks first by a substantial margin in both tree models, confirming the logistic regression finding that age is the dominant predictor. This is interpretable: older household heads in the bottom 40 percent have fewer earning years ahead, are more fixed in their income trajectory, and face higher barriers to wage renegotiation or career transitions. Age functions here as a proxy for remaining time horizon rather than a direct causal mechanism. Second, race: White (Non-Hispanic) ranks second in both Random Forest and XGBoost, identical to the logistic regression. Third, education outranks net worth in both tree models (education: rank 3 or 4; net worth: rank 6 or 8), which partially contradicts Prediction 1. The SHAP analysis in Section 3.5 examines whether this ranking reflects independent or interactive effects.

Post-Hoc Exploration: SHAP Analysis

SHapley Additive exPlanations (SHAP) were computed for the tuned XGBoost model on the full training set using the TreeExplainer. SHAP values provide model-agnostic, locally faithful feature attributions: for each household, each feature's SHAP value represents its contribution (in log-odds units) to the model's prediction, relative to the mean prediction across all households. This makes SHAP interpretations more

reliable than Gini importance for feature ranking because SHAP accounts for feature interactions and does not overweight high-cardinality features.

SHAP Summary: Feature Importance

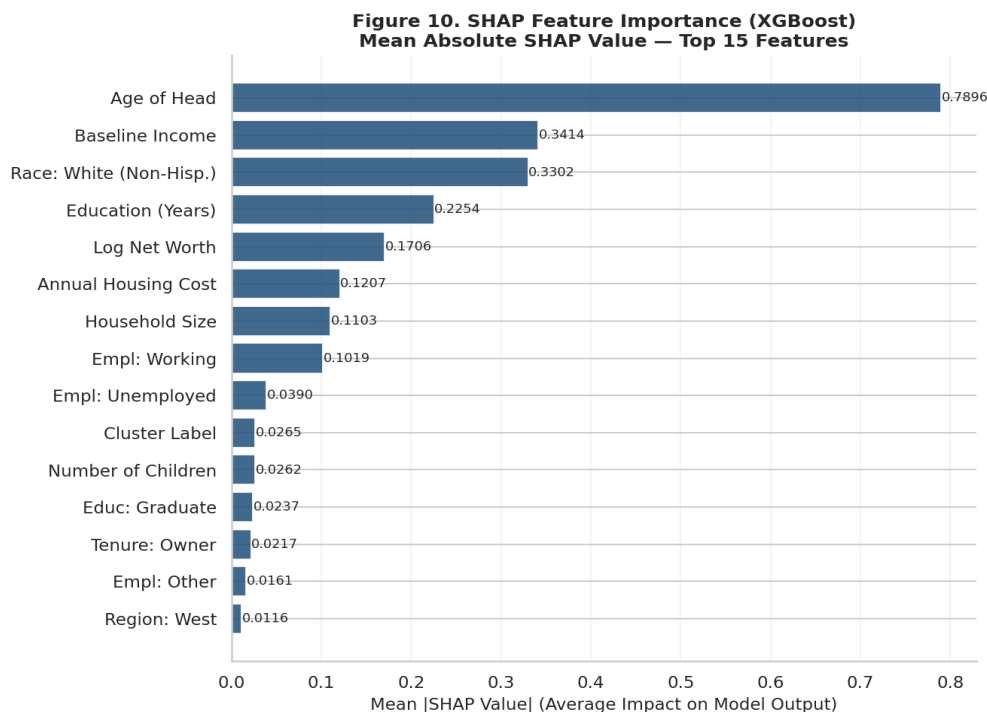


Figure 10. SHAP Feature Importance (XGBoost): Mean absolute SHAP value for the top 15 features, computed on the training set (n=1,898). Age of Head dominates with a mean |SHAP| of 0.79, more than twice the value of the next feature (Baseline Income, 0.34). Race: White (Non-Hispanic) ranks third at 0.33. Education (Years) ranks fourth at 0.23, outranking Log Net Worth (0.17).

Figure 10 confirms the feature importance rankings from Table 5 with greater precision. Age of Head dominates with a mean absolute SHAP value of 0.79, more than twice the second-ranked feature (Baseline Income, 0.34). Race: White (Non-Hispanic) ranks third (0.33), confirming that the racial mobility gap is not a linear model artifact but a genuine nonlinear signal. Education (Years) ranks fourth (0.23) and outranks Log Net Worth (0.17) in SHAP terms, partially contradicting Prediction 1. Annual housing cost ranks sixth (0.12), consistent with Prediction 2 that housing costs carry real predictive weight, though housing cost burden is not the dominant predictor the hypothesis anticipated.

SHAP Beeswarm: Direction of Effects

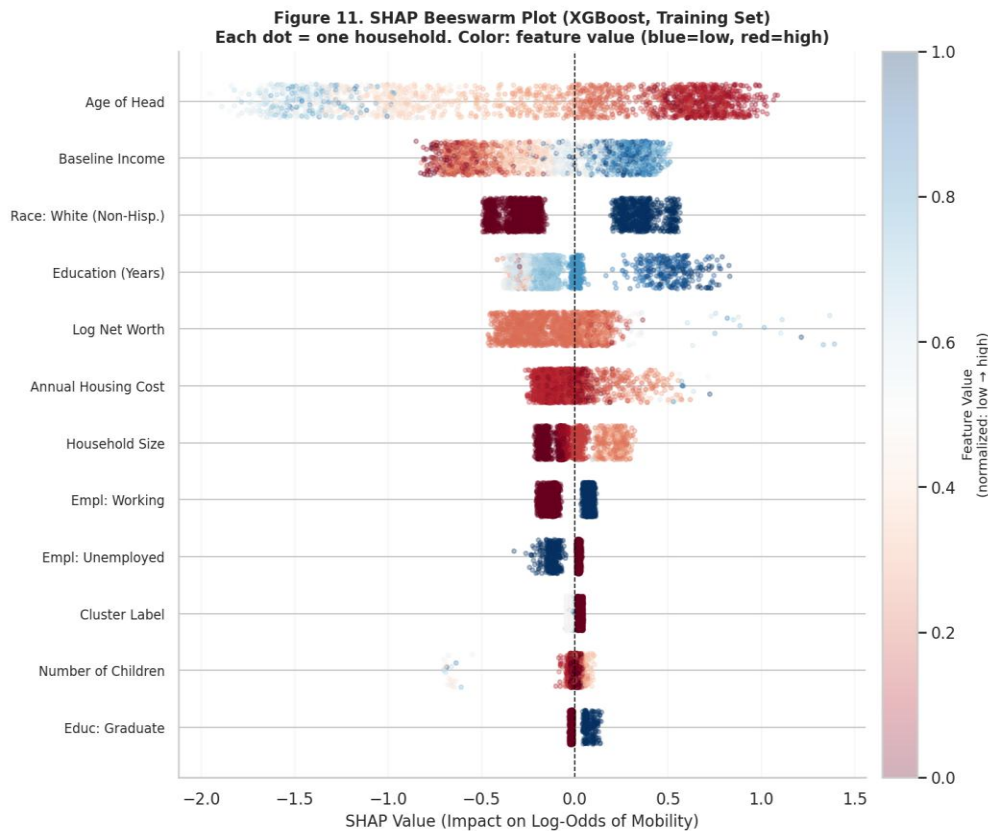


Figure 11. SHAP Beeswarm Plot (XGBoost, Training Set): Each point represents one household. Position on the x-axis shows the magnitude and direction of the SHAP value (positive = contributes to mobility prediction; negative = suppresses it). Color indicates the feature value for that household (blue = low value; red = high value).

Figure 11 reveals the direction of every feature's contribution across all training households. Age of Head shows the most dramatic pattern: households with high age (red) are concentrated in the far left of the SHAP distribution, with SHAP values extending to -2.0, meaning older household heads face a very large model penalty against mobility prediction. Households with low age (blue) cluster on the positive side, but the effect is asymmetric: the downward age penalty is substantially larger than the upward young-age boost, reflecting that older low-income households face compounding structural constraints. Race: White (Non-Hispanic) is a binary variable showing a clean split: White households receive a consistent positive SHAP contribution; non-White households receive a consistent negative contribution, with the gap representing the unexplained

racial mobility penalty. Log Net Worth shows a rightward long tail for high-wealth households, indicating that substantial assets provide a meaningful boost to mobility predictions. Annual housing cost shows predominantly negative SHAP values; higher housing costs suppress mobility predictions, consistent with Prediction 2.

SHAP Dependence: Education-by-Wealth Interaction

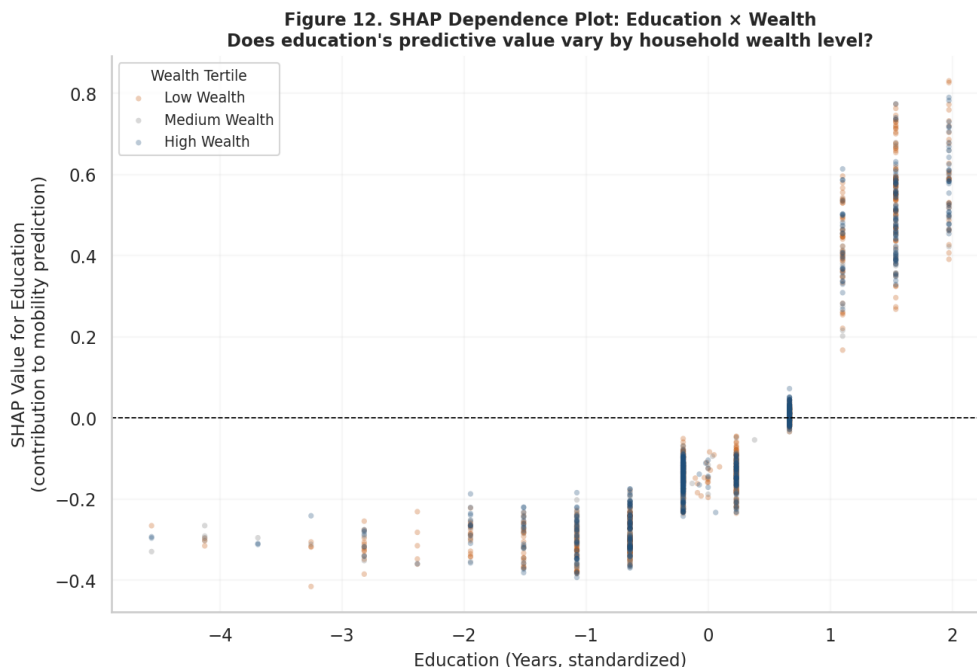


Figure 12. SHAP Dependence Plot: Education (Years) × Wealth Level. X-axis: standardized education years. Y-axis: SHAP value for education (its contribution to the mobility prediction). Points colored by wealth tertile (orange = low wealth; gray = medium wealth; blue = high wealth).

Figure 12 directly tests Prediction 4: whether education's predictive value is conditional on baseline wealth level. The result partially disconfirms the prediction. Across the lower education range (standardized values from -5 to 0, corresponding roughly to less than high school through high school diploma), SHAP values for education are predominantly negative, and the three wealth tertiles overlap substantially, meaning the model assigns a similar education SHAP penalty to low-education households regardless of their wealth level. At higher education levels (standardized values from 0.5 to 2.0, corresponding to some college through

graduate degree), large positive SHAP contributions appear across all three wealth tertiles, not concentrated in the high-wealth group.

This pattern suggests that education and wealth operate as roughly independent contributors to mobility predictions in this dataset, rather than in the strongly conditional pattern Prediction 4 anticipated. High education boosts mobility predictions for households at all wealth levels; low education suppresses predictions similarly across wealth levels. The interaction is present but modest: high-wealth households at high education levels do show slightly larger positive SHAP values than low-wealth households at equivalent education, but the difference is not the dominant pattern in the plot. This finding has policy implications: it suggests that education interventions may have real mobility benefits even for households without significant asset bases, though the moderate mean SHAP magnitude for education (0.23) relative to age (0.79) indicates the effect size is bounded.

Discussion and Next Steps

Summary of Key Findings

This project set out to answer which structural baseline conditions best predict upward income mobility for households in the bottom 40 percent of the U.S. income distribution. Three supervised classification models, logistic regression, random forest, and XGBoost, were applied to 2,375 panel-linked PSID households, with SHAP analysis providing post-hoc interpretability for the best-performing model. The answers are consistent across all three models and the SHAP analysis.

Age of household head is the dominant predictor of mobility, accounting for 32.9 percent of Random Forest importance and the highest mean absolute SHAP value (0.79)

in the XGBoost model. This finding is interpretable and consequential: older household heads in the bottom 40 percent have fewer earning years ahead, more fixed income trajectories, and fewer opportunities to retrain or advance. Age is not a direct causal mechanism that policy can address; no intervention makes people younger, but it identifies where other interventions may have the greatest marginal return: households where the head is in the middle stage of their career have more time to benefit from asset-building programs, credential completion supports, or housing cost relief.

Race is the second most consistent predictor across all three models. White (Non-Hispanic) households have 2.6 times the odds of mobility as Black/African American households in the logistic regression, and race: White ranks second in both Random Forest and XGBoost after controlling for income, wealth, education, employment, housing cost, and geography. The SHAP beeswarm confirms this is not a distributional artifact: the racial SHAP split is sharp and consistent across the training set. This gap points to unmeasured structural barriers, discrimination in hiring, differential access to credit, neighborhood quality, and intergenerational wealth transfers that are not captured by the PSID variables available in this study but that operate through race and produce the observed mobility differential.

The central hypothesis, that wealth and housing cost burden are stronger predictors than education, is partially supported. Housing cost burden is confirmed as a negative predictor (Prediction 2), and the K-Means clusters identify a distinct high-burden/low-asset household profile with substantially lower mobility rates (Prediction 3). However, in tree-based feature importance and SHAP rankings,

education consistently outranks net worth. The SHAP dependence analysis (Prediction 4) finds that education and wealth operate as roughly independent contributors rather than in a strongly conditional pattern. Education provides a real, if bounded, boost to mobility predictions across all wealth levels. The correct interpretation of these results is not that wealth is unimportant; it ranks in the top six across all models, but that in a sample where income variation is constrained by design, the education gradient is large enough to outperform wealth in global importance rankings.

Recommendations

The following recommendations are directed to policymakers and nonprofit organizations working on economic mobility and housing equity, including the Urban Institute and NLIHC as primary stakeholders, and are grounded in the model findings rather than the project's prior hypotheses

5. Direct program resources toward mid-career asset building, not just income support. Age is the dominant predictor in every model, but it functions as a time-horizon proxy: the policy implication is that households in their 30s and early 40s represent the optimal window for interventions that require time to compound, matched savings programs, homeownership pathways, and emergency fund development. Net worth consistently ranks among the top six predictors, and the SHAP beeswarm confirms it provides a meaningful positive contribution to mobility predictions for high-wealth households. Programs that convert income into assets during the critical mid-career window have the greatest structural leverage.

6. Pair housing cost relief with credential completion and target completion rather than enrollment. The model confirms that high housing cost is a negative mobility predictor across all three models. High housing cost burden reduces the capacity to save, invest, or absorb shocks, and the cluster analysis shows that high-burden households have a 14.9 percent mobility rate versus 28.0 percent for moderate-wealth households. At the same time, incomplete post-secondary credentials carry a negative model association, consistent with Putnam's (2015) finding that some college without degree completion can worsen economic outcomes relative to direct labor market entry. Programs that bundle rental assistance with completion-focused education support address both levers simultaneously.
7. Design explicitly race-conscious interventions, because race-neutral programs will not close the gap. The racial mobility gap (13.3% for Black households vs. 33.2% for White households in this sample) survives every structural control applied, income, wealth, education, employment, housing, and geography, in three separate models using different estimation strategies. Race-neutral programs operating through these levers will not close a gap that operates outside them. Policy and program design that explicitly targets the unmeasured structural barriers, equitable credit access, anti-discrimination enforcement in hiring, and wealth-building for Black households specifically, is a necessary component of any strategy aimed at closing the racial mobility gap.
8. Use the model as a targeting tool, not a verdict on individual households. The XGBoost model achieves an AUC of 0.811 and correctly identifies approximately

73 percent of households that achieve mobility. This is sufficient performance to use as a prioritization instrument, directing program outreach, eligibility screening, or resource allocation toward households with the structural profiles the model identifies as high-barrier. It is not sufficient for high-stakes individual decisions. The precision of 0.43 on the mobility class means the model produces false positives, and false negatives represent genuine mobile households that the model misses.

Future Directions

Several extensions would strengthen and expand this work. First, SHAP interaction values (`shap_values.values[:, :, :]` from SHAP's TreeExplainer for XGBoost) would provide a more formal test of the education-by-wealth interaction than the dependence plot approximation used here. The dependence plot shows the pattern; interaction values would quantify it with statistical precision. Second, the model was trained on 2011 baseline data, predating the pandemic, the 2017 Tax Cuts and Jobs Act, and substantial shifts in the housing and labor markets. Recalibration on more recent PSID waves is recommended before any deployment application. Third, the current model classifies households as mobile or not mobile relative to a single national income threshold. A gradient outcome, measuring the magnitude of income change rather than whether the threshold was crossed, would capture partial mobility and provide a more nuanced target for intervention design. Fourth, the model's features are available at the household level in the PSID but deploying it as a practical targeting tool would require mapping to administrative data sources (SNAP enrollment records, housing voucher

waitlists, community college enrollment) that can provide similar baseline profiles for real program participants.

Caveats and Threats to Validity

Internal Threats

The most significant internal threat is omitted variable bias. The racial mobility gap survives all measured controls, but unmeasured variables, credit scores, neighborhood institutional quality, family wealth transfers not captured by PSID net worth, and discrimination histories could explain part of the residual race coefficient. The model cannot distinguish between race as a direct predictor and race as a proxy for these unmeasured structural conditions. Additionally, the PSID's measurement of wealth uses imputed values from the wealth supplement module; imputation introduces measurement error that may attenuate wealth's predictive power relative to its true effect.

The K-Means cluster labels are a derived feature, and although the leakage audit (AUC = 0.61) confirmed that the clusters carry genuine structural signal rather than test-set contamination, the cluster labels are not interpretable in the same way as measured features. Households assigned to the same cluster share similar structural profiles, but cluster membership is sensitive to the choice of k and the set of clustering features.

External Threats

The PSID oversamples low-income households, which improves statistical power for this study but means that the 21.5 percent overall mobility rate is not a nationally representative estimate. Mobility rates would differ in a population-

representative sample. The analytic cohort observed a decade of income change from 2011 to 2021, a period that includes significant labor market tightening, the COVID-19 pandemic (which affected 2020 income measurement in the 2021 wave), and housing market disruption. These contextual forces may have affected income mobility outcomes in ways that are not separable from the structural predictors in the model. Findings from this cohort should be applied cautiously to households observed under different macroeconomic conditions.

Code Availability

All code for this project is available in the public GitHub repository at:

<https://github.com/JCharles8/mobility-predictors>

The repository contains: the Jupyter notebook (Code.ipynb) with all EDA, preprocessing, modeling, and SHAP analysis cells; the rendered HTML version of the notebook; the `psid_panel.csv` analytic dataset; output figures; and a `README.md` with authorship information, project overview, dependency list, and run instructions. The repository will remain static for at least 12 months following submission.

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Appendix A: Data Dictionary / Codebook

The following table describes all variables used in the analysis, including PSID source variable codes, operational definitions, measurement levels, and coding decisions.

Variable Name	PSID Code(s)	Definition	Type	Coding Notes
income_2010	ER52343	Total family income, calendar year 2010	Continuous (\$)	Restricted to ≤40th percentile (\$37,190). N=2,375.
wealth_no_equity	ER52392	Imputed net worth excluding home equity, 2011 wealth supplement	Continuous (\$)	Primary wealth predictor. Skew=16.9. Sign-preserving log applied. Values ±9,999,999 recoded missing.
wealth_with_equity	ER52394	Imputed net worth including home equity, 2011 wealth supplement	Continuous (\$)	Collinear with excl. equity (r=0.89). Dropped from models; retained in EDA.
age_head	ER47317	Age in years of household head at 2011 interview	Continuous (years)	Values ≥98 recoded missing. Range: 17–96. Median: 36.
hh_size	ER47316	Total number of persons in the family unit	Continuous (count)	Includes head, partner, children, and other FU members.
num_children	ER47320	Number of children in family unit at 2011 interview	Continuous (count)	Biological, adoptive, and stepchildren under 18.
education_years	ER52405	Highest grade of schooling completed by household head	Ordinal	≤11=<HS; 12=HS; 13–14=Some College; 15=BA; 16–17=Grad.

Variable Name	PSID Code(s)	Definition	Type	Coding Notes
employment_status	ER47448	Employment status of household head (first mention)	Nominal (5 cats.)	Original 8 codes collapsed: 1=Working; 2=Temp Laid Off; 3=Unemployed; 4–5=Not in LF; 6–8=Other.
family_structure	ER52407	Marital/partnership status of household head	Binary (nominal)	Codes 1–3 = Married/Partnered; codes 4–7 = Single/Unpartnered.
race_ethnicity	ER51904, ER51903	Race and Hispanic origin of household head	Nominal (4 cats.)	Hispanic origin takes precedence. White (Non-Hispanic), Black/AA, Hispanic, Other.
geography	ER52398	PSID Census region of household residence at 2011 interview	Nominal (4 cats.)	1=Northeast; 2=Midwest; 3=South; 4–5=West.
housing_tenure	ER47329	Whether household owns, rents, or has other arrangement	Nominal (3 cats.)	1=Owner; 5=Renter; 8=Neither. Neither group lacks housing cost data.
monthly_housing_cost	ER47350, ER47371, ER47387	Monthly housing expenditure	Continuous (\$)	Owner: sum of primary and secondary mortgage payments. Renter: monthly rent. Missing for Neither tenure.
annual_housing_cost	Derived	Monthly housing cost $\times$ 12	Continuous (\$)	Missing for 606 Neither-tenure households. Binary

Variable Name	PSID Code(s)	Definition	Type	Coding Notes
				missing indicator retained before imputation.
housing_cost_burden	Derived	Annual housing cost / annual income	Continuous (ratio)	Clipped at 2.0. Missing for 606 households. Used in EDA; components entered separately in models.
housing_cost_missing	Derived	Binary indicator: housing cost missing	Binary	Constructed before MICE imputation to preserve Neither-tenure signal.
log_net_worth	Derived	sign_preserving_log(wealth_no_equity)	Continuous	Reduces skew from 17.0 to ~4.5. Handles negative net worth values.
cluster_label	Derived	K-Means cluster assignment (k=3)	Nominal (0,1,2)	Fit on training set only. Leakage audit AUC = 0.61 (PASS). Cluster 0: 14.9% mobility; Cluster 1: 28.0%; Cluster 2: 25.4%.
above_median_2021 (TARGET)	ER81775 (2021)	Did household income in 2020 exceed the national median (\$67,521)?	Binary (0/1)	Mobility rate: 21.5%. Linked via Cross-Year Individual File (IND2023ER).
Panel Linkage IDs	ER34101, ER34901	2011 and 2021 family interview numbers from Cross-Year Individual file	Identifier	Used to merge FAM2011ER to FAM2021ER. Sequence number ER34102=1 identifies household heads in 2011.

Note: All baseline variables are drawn from the 2011 PSID Family File (FAM2011ER) unless otherwise noted. Panel linkage uses the 2023 Cross-Year Individual File (IND2023ER). Target variable income is from the 2021 Family File (FAM2021ER).

Appendix B: Logistic Regression Coefficient Table

Table B1 presents the top 10 logistic regression coefficients by absolute magnitude after L2 regularization, from the standardized feature matrix.

Feature	Coefficient	Odds Ratio	Rank	Direction
age_head	-1.090	0.336	1	Negative
race: White (Non-Hispanic)	0.959	2.609	2	Positive
race: Hispanic	0.664	1.943	3	Positive
log_net_worth	0.434	1.544	4	Positive
employment: Temp Laid Off	-0.410	0.663	5	Negative
housing_tenure: Owner	0.396	1.486	6	Positive
income_2010	0.382	1.465	7	Positive
education_years	0.363	1.438	8	Positive
education: Some College / AA	-0.361	0.697	9	Negative
employment: Unemployed	-0.358	0.699	10	Negative

Table B1. Top 10 logistic regression coefficients by absolute magnitude after L2 regularization (C=1.0). Coefficients are from the standardized feature matrix, making relative magnitudes comparable. Odds ratios = $\exp(\text{coefficient})$. Reference categories: Black/African American (race), Working (employment), Renter (housing tenure), Less than HS (education).